Enterprise Campus Design

As discussed in Chapter 2, the Enterprise Campus functional area is divided into the following modules:

- Campus Infrastructure—This module includes three layers:
  - The Building Access layer
  - The Building Distribution layer
  - The Campus Core layer
- Server Farm
- Edge Distribution (optional)

This section discusses the design of each of the layers and modules within the Enterprise Campus and identifies best practices related to the design of each.

Enterprise Campus Requirements

As shown in Table 4-3, each Enterprise Campus module has different requirements. For example, this table illustrates how modules located closer to the users require a higher degree of scalability so that the Campus network can be expanded in the future without redesigning the complete network. For example, adding new workstations to a network should result in neither high investment cost nor performance degradations.

Table 4-3: Enterprise Campus Design Requirements
Open table as spreadsheet

Requirement	Building Access	Building Distribution	Campus Core	Server Farm	Edge Distribution
Technology	Data link layer or multilayer switched	Multilayer switched	Multilayer switched	Multilayer switched	Multilayer switched
Scalability	High	Medium	Low	Medium	Low
High availability	Medium	Medium	High	High	Medium
Performance	Medium	Medium	High	High	Medium
Cost per port	Low	Medium	High	High	Medium

End users (in the Building Access layer) usually do not require high performance or high availability, but these features are crucial to the Campus Core layer and the Server Farm module.

The price per port increases with increased performance and availability. The Campus Core and Server Farm require a guarantee of higher throughput so they can handle all traffic flows and not introduce additional delays or drops to the network traffic.

The Edge Distribution module does not require the same performance as in the Campus Core. However, it can require other features and functionalities that increase the overall cost.

Building Access Layer Design Considerations

When implementing the campus infrastructure's Building Access layer, consider the following questions:

- How many users or host ports are currently required in the wiring closet, and how many will it require in the future? Should the switches be fixed or modular configuration?
- How many ports are available for end-user connectivity at the walls of the buildings?
- How many access switches are not located in wiring closets?
- What cabling is currently available in the wiring closet, and what cabling options exist for uplink connectivity?
- What data link layer performance does the node need?
- What level of redundancy is needed?
- What is the required link capacity to the Building Distribution layer switches?
- How will VLANs and STP be deployed? Will there be a single VLAN, or several VLANs per access switch? Will the VLANs on the switch be unique or spread across multiple switches? The latter design was common a few years ago, but today end-to-end VLANs (also called *campuswide VLANs*) are not desirable.
- Are additional features, such as port security, multicast traffic management, and QoS (such as traffic classification based on ports), required?

Based on the answers to these questions, select the devices that satisfy the Building Access layer's requirements. The Building Access layer should maintain the simplicity of traditional LAN switching, with the support of basic network intelligent services and business applications.

Key Point

The following are best-practice recommendations for optimal Building Access layer design:

- Manage VLANs and STP
- Manage trunks between switches
- Manage default Port Aggregation Protocol (PAgP) settings
- Consider implementing routing

These recommendations are described in the following sections.

Managing VLANs and STP

This section details best-practice recommendations related to VLANs and STP.

Limit VLANs to a Single Wiring Closet Whenever Possible

As a best practice, limit VLANs to a single wiring closet whenever possible.

NoteCisco (and other vendors) use the term *local VLAN* to refer to a VLAN that is limited to a single wiring closet.

Avoid Using STP if Possible

STP is defined in IEEE 802.1d. Avoid requiring any type of STP (including Rapid STP [RSTP]) by design for the most deterministic and highly available network topology that is predictable and bounded and has reliably tuned convergence.

For example, the behavior of Layer 2 environments (using STP) and Layer 3 environments (using a routing protocol) are different under "soft failure" conditions, when keepalive messages are lost. In an STP environment, if bridge protocol data units (BPDU) are lost, the network fails in an "open" state, forwarding traffic with unknown destinations on all ports, potentially causing broadcast storms.

In contrast, routing environments fail "closed," dropping routing neighbor relationships, breaking connectivity, and isolating the soft failed devices.

Another reason to avoid using STP is for load balancing: If there are two redundant links, STP by default uses only one of the links, while routing protocols by default use both.

If STP Is Required, Use RSTP with Per-VLAN Spanning Tree Plus

Cisco developed Per-VLAN Spanning Tree (PVST) so that switches can have one instance of STP running per VLAN, allowing redundant physical links within the network to be used for different VLANs and thus reducing the load on individual links. PVST works only over ISL trunks. However, Cisco extended this functionality for 802.1Q trunks with the Per-VLAN Spanning Tree Plus (PVST+) protocol. Before this became available, 802.1Q trunks supported only Common Spanning Tree (CST), with one instance of STP running for all VLANs.

Multiple-Instance STP (MISTP) is an IEEE standard (802.1s) that uses RSTP and allows several VLANs to be grouped into a single spanning-tree instance. Each instance is independent of the other instances so that a link can forward for one group of VLANs while blocking for other VLANs. MISTP therefore allows traffic to be shared across all the links in the network, but it reduces the number of STP instances that would be required if PVST/PVST+ were implemented.

RSTP is defined by IEEE 802.1w. RPVST+ is a Cisco enhancement of RSTP. As a best practice, if STP must be used, use RPVST+.

NoteWhen Cisco documentation refers to implementing RSTP, it is referring to RPVST+.

The Cisco RPVST+ implementation is far superior to 802.1d STP and even PVST+ from a convergence perspective. It greatly improves the convergence times for any VLAN on which a link comes up, and it greatly improves the convergence time compared to BackboneFast (as described in the next section) for any indirect link failures.

Two other STP-related recommendations are as follows:

- If a network includes non-Cisco switches, isolate the different STP domains with Layer 3 routing to avoid STP compatibility issues.
- Even if the recommended design does not depend on STP to resolve link or node failure events, use STP in Layer 2 designs to protect against user-side loops. A loop can be introduced on the user-facing access layer ports in many ways, such as wiring mistakes, misconfigured end stations, or malicious users. STP is required to ensure a loop-free topology and to protect the rest of the network from problems created in the access layer.

NoteSome security personnel have recommended disabling STP at the network edge. Cisco does not recommend this practice, however, because the risk of lost connectivity without STP is far greater than any STP information that might be revealed.

The Cisco STP Toolkit

The Cisco STP toolkit provides tools to better manage STP when RSTP+ is not available:

- **PortFast:** Used for ports to which end-user stations or servers are directly connected. When PortFast is enabled, there is no delay in passing traffic, because the switch immediately puts the port in STP forwarding state, skipping the listening and learning states. Two additional measures that prevent potential STP loops are associated with the PortFast feature:
    - **BPDU Guard**: PortFast transitions the port into the STP forwarding state immediately on linkup. Because the port still participates in STP, the potential for an STP loop exists if some device attached to that port also runs STP. The BPDU Guard feature enforces the STP domain borders and keeps the active topology predictable. If the port receives a BPDU, the port is transitioned into *errdisable state* (meaning that it was disabled due to an error), and an error message is reported.

      Note Additional information on the errdisable state is available in *Recovering from errDisable Port State on the CatOS Platforms*, at http://www.cisco.com/en/US/tech/tk389/tk214/technologies_tech_note09186a0080093dcb.shtml.

    - **BPDU Filtering**: This feature blocks PortFast-enabled, nontrunk ports from transmitting BPDUs. STP does not run on these ports. BPDU filtering is not recommended, because it effectively disables STP at the edge and can lead to STP loops.
- **UplinkFast:**If the link on a switch to the root switch goes down and the blocked link is directly connected to the same switch, UplinkFast enables the switch to put a redundant port (path) into the forwarding state immediately, typically resulting in convergence of 3 to 5 seconds after a link failure.
- **BackboneFast:** If a link on the way to the root switch fails but is *not* directly connected to the same switch (in other words, it is an indirect failure), BackboneFast reduces the convergence time by max_age (which is 20 seconds by default), from 50 seconds to approximately 30 seconds. When this feature is used, it must be enabled on all switches in the STP domain.
- **STP Loop Guard:** When one of the blocking ports in a physically redundant topology stops receiving BPDUs, usually STP creates a potential loop by moving the port to forwarding state. With the STP Loop Guard feature enabled, and if a blocking port no longer receives BPDUs, that port is moved into the STP loop-inconsistent blocking state instead of the listening/learning/forwarding state. This feature avoids loops in the network that result from unidirectional or other software failures.
- **RootGuard:** The RootGuard feature prevents external switches from becoming the root. RootGuard should be enabled on all ports where the root bridge should *not* appear; this feature ensures that the port on which RootGuard is enabled is the designated port. If a *superior* BPDU (a BPDU with a lower bridge ID than that of the current root bridge) is received on a RootGuard-enabled port, the port is placed in a root-inconsistent state—the equivalent of the listening state.
- **BPDU Skew Detection:** This feature allows the switch to keep track of late-arriving BPDUs (by default, BPDUs are sent every 2 seconds) and notify the administrator via Syslog messages. Skew detection generates a report for every port on which BPDU has ever arrived late (this is known as *skewed* arrival). Report messages are rate-limited (one message every 60 seconds) to protect the CPU.
- **Unidirectional Link Detection (UDLD):** A unidirectional link occurs whenever traffic transmitted by the local switch over a link is received by the neighbor but traffic transmitted from the neighbor is not received by the local device. If the STP process that runs on the switch with a blocking port stops receiving BPDUs from its upstream (designated) switch on that port, STP eventually ages out the STP information for this port and moves it to the forwarding state. If the link is unidirectional, this action would create an STP loop. UDLD is a Layer 2 protocol that works with the Layer 1 mechanisms to determine a link's physical status. If the port does not see its own device/port ID in the incoming UDLD packets for a specific duration, the link is considered unidirectional from the Layer 2 perspective. After UDLD detects the unidirectional link, the respective port is disabled, and an error message is generated.

NotePortFast, Loop Guard, RootGuard, and BPDU Guard are also supported for RPVST+.

As an example of the use of these features, consider when a switch running a version of STP is introduced into an operating network. This might not always cause a problem, such as when the switch is connected in a conference room to temporarily provide additional ports for connectivity. However, sometimes this is undesirable, such as when the switch that is added has been configured to become the STP root for the VLANs to which it is attached. BDPU Guard and RootGuard are tools that can protect against these situations. BDPU Guard requires operator intervention if an unauthorized switch is connected to the network, and RootGuard protects against a switch configured in a way that would cause STP to reconverge when it is being connected to the network.

Managing Trunks Between Switches

Trunks are typically deployed on the interconnection between the Building Access and Building Distribution layers. There are several best practices to implement with regard to trunks.

Trunk Mode and Encapsulation

As a best practice when configuring trunks, set Dynamic Trunking Protocol (DTP) to **desirable** on one side and **desirable** (with the **negotiate** option) one the other side to support DTP protocol (encapsulation) negotiation.

NoteAlthough turning DTP to **on** and **on** with the **no negotiate** option could save seconds of outage when restoring a failed link or node, with this configuration DTP does not actively monitor the state of the trunk, and a misconfigured trunk is not easily identified.

NoteThe specific commands used to configure trunking vary; refer to your switch's documentation for details.

Manually Pruning VLANs

Another best practice is to manually prune unused VLANs from trunked interfaces to avoid broadcast propagation. Cisco recommends not using automatic VLAN pruning; manual pruning provides stricter control. As mentioned, campuswide or access layer–wide VLANs are no longer recommended, so VLAN pruning is less of an issue than it used to be.

VTP Transparent Mode

VTP transparent mode should be used as a best practice because hierarchical networks have little need for a shared common VLAN database. Using VTP transparent mode decreases the potential for operational error.

Trunking on Ports

Trunking should be disabled on ports to which hosts will be attached so that host devices do not need to negotiate trunk status. This practice speeds up PortFast and is a security measure to prevent VLAN hopping.

Managing Default PAgP Settings

Fast EtherChannel and Gigabit EtherChannel solutions group several parallel links between LAN switches into a channel that is seen as a single link from the Layer 2 perspective. Two protocols handle automatic EtherChannel formation: PAgP, which is Cisco-proprietary, and the Link Aggregation Control Protocol (LACP), which is standardized and defined in IEEE 802.3ad.

When connecting a Cisco IOS software device to a Catalyst operating system device using PAgP, make sure that the PAgP settings used to establish EtherChannels are coordinated; the defaults are different for a Cisco IOS software device and a Catalyst operating system device. As a best practice, Catalyst operating system devices should have PAgP set to **off** when connecting to a Cisco IOS software device if EtherChannels are not configured. If EtherChannel/PAgP is used, set both sides of the interconnection to **desirable**.

Implementing Routing in the Building Access Layer

Although not as widely deployed in the Building Access layer, a routing protocol, such as EIGRP, when properly tuned, can achieve better convergence results than Layer 2 and Layer 3 boundary hierarchical designs that rely on STP. However, adding routing does result in some additional complexities, including uplink IP addressing and subnetting, and loss of flexibility.

Figure 4-9 illustrates a sample network with Layer 3 routing in both the Building Access and Building Distribution layers. In this figure, equal-cost Layer 3 load balancing is performed on all links (although EIGRP could perform unequal-cost load balancing). STP is not run, and a first-hop redundancy protocol (such as Hot Standby Router Protocol [HSRP]) is not required. VLANs cannot span across the multilayer switch.

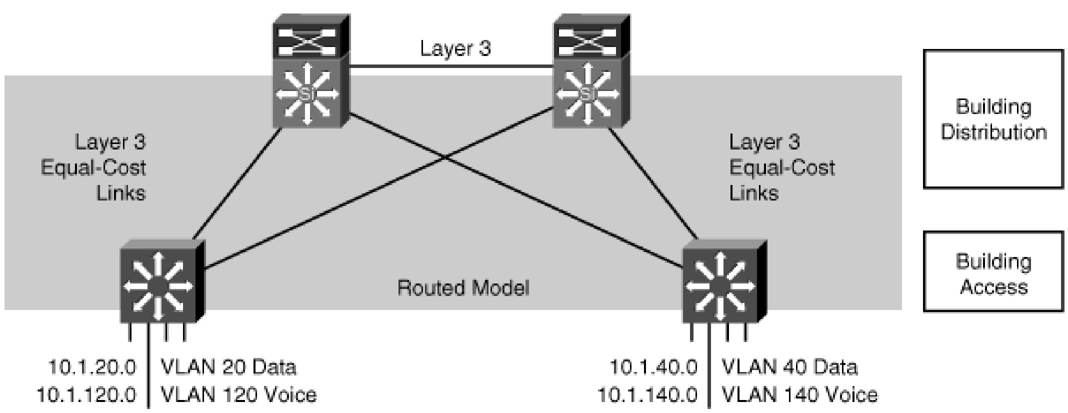

Figure 4-9: Layer 3 Access-to-Distribution Layer Interconnection

Note HSRP and other first-hop redundancy protocols are discussed in the "Using First-Hop Redundancy Protocols" section.

Building Distribution Layer Design Considerations

The Building Distribution layer aggregates the Building Access layer, segments workgroups, and isolates segments from failures and broadcast storms. This layer implements many policies based on access lists and QoS settings. The Building Distribution layer can protect the Campus Core network from any impact of Building Access layer problems by implementing all the organization's policies.

When implementing the Building Distribution layer, consider the following questions:

- How many devices will each Building Distribution switch handle?
- What type and level of redundancy are required?
- How many uplinks are needed?
- What speed do the uplinks need to be to the building core switches?
- What cabling is currently available in the wiring closet, and what cabling options exist for uplink connectivity?
- As network services are introduced, can the network continue to deliver high performance for all its applications, such as video on demand, IP multicast, or IP telephony?

The network designer must pay special attention to the following network characteristics:

- **Performance:** Building Distribution switches should provide wire-speed performance on all ports. This feature is important because of Building Access layer aggregation on one side and high-speed connectivity of the Campus Core module on the other side. Future expansions with additional ports or modules can result in an overloaded switch if it is not selected properly.
- **Redundancy:** Redundant Building Distribution layer switches and redundant connections to the Campus Core should be implemented. Using equal-cost redundant connections to the core supports fast convergence and avoids routing black holes. Network bandwidth and capacity should be engineered to withstand node or link failure.

  When redundant switches cannot be implemented in the Campus Core and Building Distribution layers, redundant supervisors and the Stateful Switchover (SSO) and Nonstop Forwarding (NSF) technologies can provide significant resiliency improvements. These technologies result in 1 to 3 seconds of outage in a failover, which is less than the time needed to replace a supervisor and recover its configuration. Depending on the switch platform, full-image In Service Software Upgrade (ISSU) technology might be available such that the complete Cisco IOS software image can be upgraded without taking the switch or network out of service, maximizing network availability.

- **Infrastructure services:**Building Distribution switches should not only support fast multilayer switching, but should also incorporate network services such as high availability, QoS, security, and policy enforcement.

  Expanding and/or reconfiguring distribution layer devices must be easy and efficient. These devices must support the required management features.

With the correct selection of Building Distribution layer switches, the network designer can easily add new Building Access modules.

Key Point

Multilayer switches are usually preferred as the Building Distribution layer switches, because this layer must usually support network services, such as QoS and traffic filtering.

Key Point

The following are best-practice recommendations for optimal Building Distribution layer design:

- Use first-hop redundancy protocols.
- Deploy Layer 3 routing protocols between the Building Distribution switches and Campus Core switches.
- If required, Building Distribution switches should support VLANs that span multiple Building Access layer switches.

The following sections describe these recommendations.

Using First-Hop Redundancy Protocols

If Layer 2 is used between the Building Access switch and the Building Distribution switch, convergence time when a link or node fails depends on default gateway redundancy and failover time. Building Distribution switches typically provide first-hop redundancy (default gateway redundancy) using HSRP, Gateway Load-Balancing Protocol (GLBP), or Virtual Router Redundancy Protocol (VRRP).

This redundancy allows a network to recover from the failure of the device acting as the default gateway for end nodes on a physical segment. Uplink tracking should also be implemented with the first-hop redundancy protocol.

HSRP or GLBP timers can be reliably tuned to achieve subsecond (800 to 900 ms) convergence for link or node failure in the boundary between Layer 2 and Layer 3 in the Building Distribution layer.

In Cisco deployments, HSRP is typically used as the default gateway redundancy protocol. VRRP is an Internet Engineering Task Force (IETF) standards-based method of providing default gateway redundancy. More deployments are starting to use GLBP because it supports load balancing on the uplinks from the access layer to the distribution layer, as well as first-hop redundancy and failure protection.

As shown in Figure 4-10, this model supports a recommended Layer 3 point-to-point interconnection between distribution switches.

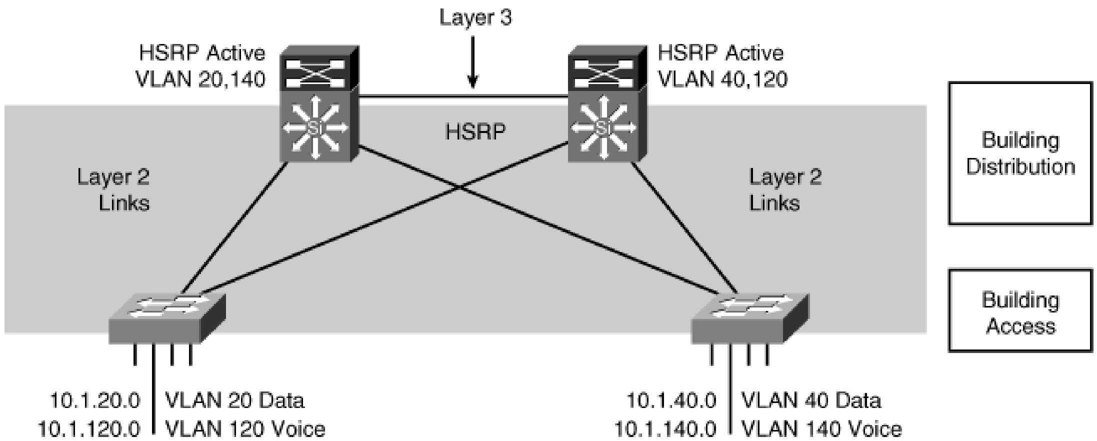

Figure 4-10: Layer 3 Distribution Switch Interconnection

No VLANs span the Building Access layer switches across the distribution switches, so from an STP perspective, both access layer uplinks are forwarding, and no STP convergence is required if uplink failure occurs. The only convergence dependencies are the default gateway and return path route selection across the Layer 3 distribution-to-distribution link.

NoteNotice in Figure 4-10 that the Layer 2 VLAN number is mapped to the Layer 3 subnet for ease of management.

If Layer 3 is used to the Building Access switch, the default gateway is at the multilayer Building Access switch, and a first-hop redundancy protocol is not needed.

Deploying Layer 3 Routing Protocols Between Building Distribution and Campus Core Switches

Routing protocols between the Building Distribution switches and the Campus Core switches support fast, deterministic convergence for the distribution layer across redundant links.

Convergence based on the up or down state of a point-to-point physical link is faster than timer-based nondeterministic convergence. Instead of indirect neighbor or route loss detection using hellos and dead timers, physical link loss indicates that a path is unusable; all traffic is rerouted to the alternative equal-cost path.

For optimum distribution-to-core layer convergence, build redundant *triangles*, not *squares*, to take advantage of equal-cost redundant paths for the best deterministic convergence. Figure 4-11 illustrates the difference.

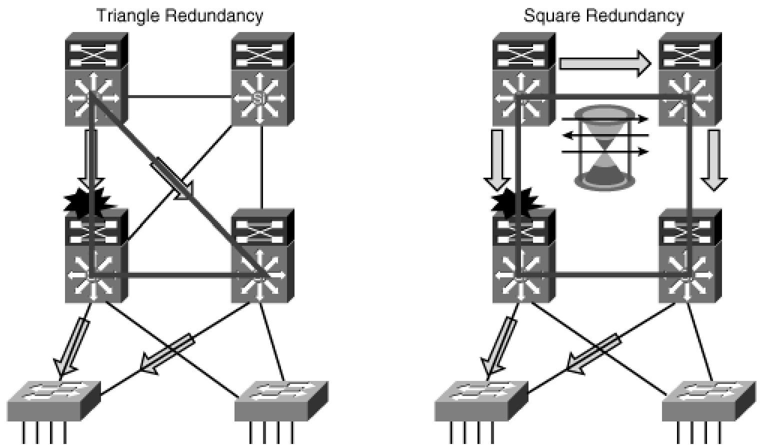

Figure 4-11: Redundant Triangles Versus Redundant Squares

On the left of Figure 4-11, the multilayer switches are connected redundantly with a triangle of links that have Layer 3 equal costs. Because the links have equal costs, they appear in the routing table (and by default will be used for load balancing). If one of the links or distribution layer devices fails, convergence is extremely fast, because the failure is detected in hardware and there is no need for the routing protocol to recalculate a new path; it just continues to use one of the paths already in its routing table. In contrast, on the right of Figure 4-11, only one path is active by default, and link or device failure requires the routing protocol to recalculate a new route to converge.

Other related recommended practices are as follows:

- Establish routing protocol peer relationships only on links that you want to use as transit links.
- Summarize routes from the Building Distribution layer into the Campus Core layer.

Supporting VLANs That Span Multiple Building Access Layer Switches

In a less-than-optimal design where VLANs span multiple Building Access layer switches, the Building Distribution switches must be linked by a Layer 2 connection, or the Building Access layer switches must be connected via trunks.

This design is more complex than when the Building Distribution switches are interconnected with Layer 3. STP convergence is required if an uplink failure occurs.

As shown in Figure 4-12, the following are recommendations for use in this (suboptimal) design:

- Use RPVST+ as the version of STP.
- Provide a Layer 2 link between the two Building Distribution switches to avoid unexpected traffic paths and multiple convergence events.
- If you choose to load-balance VLANs across uplinks, be sure to place the HSRP primary and the RPVST+ root on the same Building Distribution layer switch to avoid using the interdistribution switch link for transit.

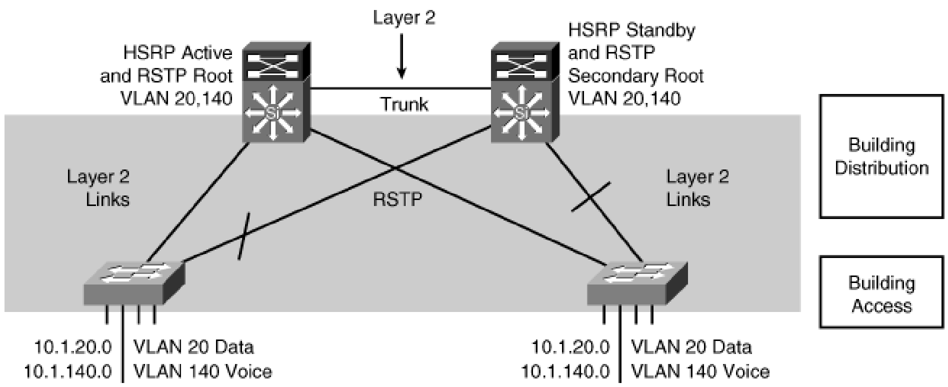

Figure 4-12: Layer 2 Building Distribution Switch Interconnection

Campus Core Design Considerations

Low price per port and high port density can govern switch choice for wiring closet environments, but high-performance wire-rate multilayer switching drives the Campus Core design.

Using Campus Core switches reduces the number of connections between the Building Distribution layer switches and simplifies the integration of the Server Farm module and Enterprise Edge modules. Campus Core switches are primarily focused on wire-speed forwarding on all interfaces and are differentiated by the level of performance achieved per port rather than by high port densities.

Key Point

As a recommended practice, deploy a dedicated Campus Core layer to connect three or more buildings in the Enterprise Campus, or four or more pairs of Building Distribution switches in a very large campus.

Campus Core switches are typically multilayer switches.

Using a Campus Core makes scaling the network easier. For example, with a Campus Core, new Building Distribution switches only need connectivity to the core rather than full-mesh connectivity to all other Building Distribution switches.

NoteNot all campus implementations need a Campus Core. As discussed in the upcoming “Small and Medium Campus Design Options” section, the Campus Core and Building Distribution layers can be combined at the Building Distribution layer in a smaller campus.

Issues to consider in a Campus Core layer design include the following:

- The performance needed in the Campus Core network.
- The number of high-capacity ports for Building Distribution layer aggregation and connection to the Server Farm module or Enterprise Edge modules.
- High availability and redundancy requirements. To provide adequate redundancy, at least two separate switches (ideally located in different buildings) should be deployed.

Another Campus Core consideration is Enterprise Edge and WAN connectivity. For many organizations, the Campus Core provides Enterprise Edge and WAN connectivity through Edge Distribution switches connected to the core. However, for large enterprises with a data center, the Enterprise Edge and WAN connectivity are aggregated at the data center module.

Typically, the Campus Core switches should deliver high-performance, multilayer switching solutions for the Enterprise Campus and should address requirements for the following:

- Gigabit density
- Data and voice integration
- LAN, WAN, and metropolitan area network (MAN) convergence
- Scalability
- High availability
- Intelligent multilayer switching in the Campus Core, and to the Building Distribution and Server Farm environments

Large Campus Design

For a large campus, the most flexible and scalable Campus Core layer consists of dual multilayer switches, as illustrated in Figure 4-13.

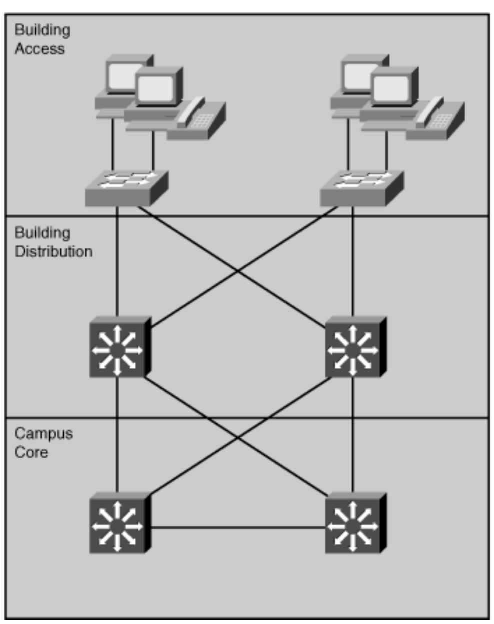

Figure 4-13: Large Campus Multilayer Switched Campus Core Design

Multilayer-switched Campus Core layers have several best-practice features:

- **Reduced multilayer switch peering (routing adjacencies):** Each multilayer Building Distribution switch connects to only two multilayer Campus Core switches, using a redundant triangle configuration. This implementation simplifies any-to-any connectivity between Building Distribution and Campus Core switches and is scalable to an arbitrarily large size. It also supports redundancy and load sharing.
- **Topology with no spanning-tree loops:** No STP activity exists in the Campus Core or on the Building Distribution links to the Campus Core layer, because all the links are Layer 3 (routed) links. Arbitrary topologies are supported by the routing protocol used in the Campus Core layer. Because the core is routed, it also provides multicast and broadcast control.
- **Improved network infrastructure services support:** Multilayer Campus Core switches provide better support for intelligent network services than data link layer core switches could support.

This design maintains two equal-cost paths to every destination network. Thus, recovery from any link failure is fast and load sharing is possible, resulting in higher throughput in the Campus Core layer.

One of the main considerations when using multilayer switches in the Campus Core is switching performance. Multilayer switching requires more sophisticated devices for high-speed packet routing. Modern Layer 3 switches support routing in the hardware, even though the hardware might not support all the features. If the hardware does not support a selected feature, it must be performed in software; this can dramatically reduce the data transfer. For example, access lists might not be processed in the hardware if they have too many entries, resulting in switch performance degradation.

Small and Medium Campus Design Options

A small campus (or large branch) network might have fewer than 200 end devices, and the network servers and workstations might be connected to the same wiring closet. Because switches in a small campus network design may not require high-end switching performance or much scaling capability, in many cases, the Campus Core and Building Distribution layers can be combined into a single layer, as illustrated on the left of Figure 4-14. This design can scale to only a few Building Access layer switches. A low-end multilayer switch provides routing services closer to the end user when multiple VLANs exist. For a very small office, one low-end multilayer switch may support the LAN access requirements for the entire office.

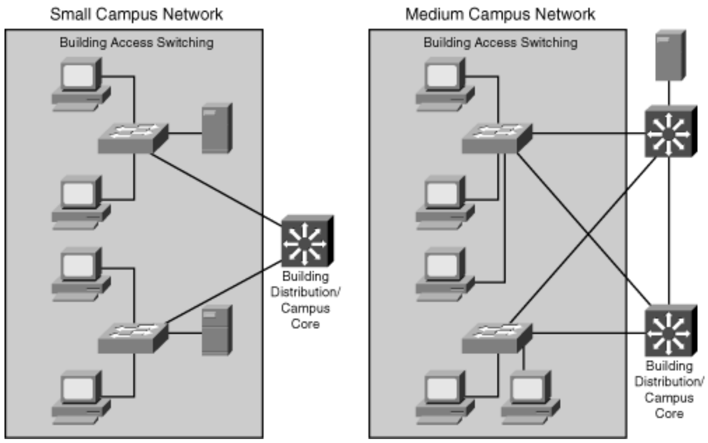

Figure 4-14: Small and Medium Campus Design Options

For a medium-sized campus with 200 to 1000 end devices, the network infrastructure typically consists of Building Access layer switches with uplinks to Building Distribution/Campus Core multilayer switches that can support the performance requirements of a medium-sized campus network. If redundancy is required, redundant multilayer switches connect to the Building Access switches, providing full link redundancy, as illustrated on the right of Figure 4-14.

NoteBranch and teleworker infrastructure considerations are described further in Chapter 5.

Edge Distribution at the Campus Core

As mentioned in Chapter 3, the Enterprise Edge modules connect to the Campus Core directly or through an optional Edge Distribution module, as illustrated in Figure 4-15.

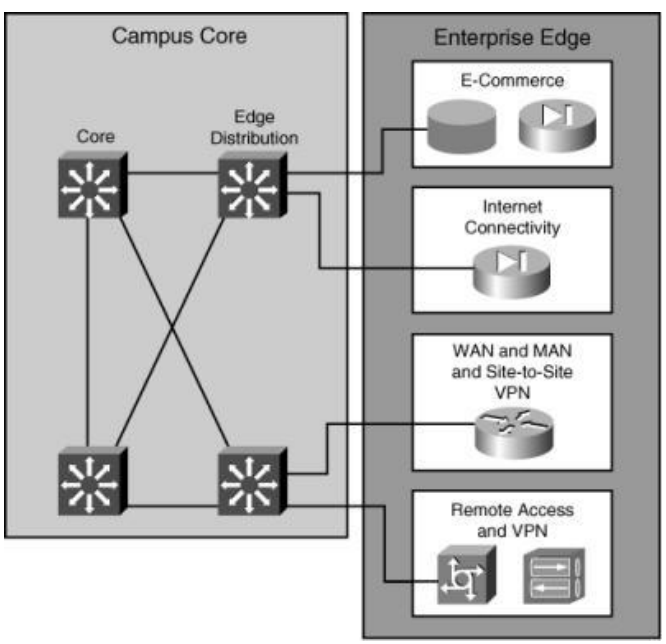

Figure 4-15: Edge Distribution Design

The Edge Distribution multilayer switches filter and route traffic into the Campus Core, aggregate Enterprise Edge connectivity, and provide advanced services.

Switching speed is not as important as security in the Edge Distribution module, which isolates and controls access to devices that are located in the Enterprise Edge modules (for example, servers in an E-commerce module or public servers in an Internet Connectivity module). These servers are closer to the external users and therefore introduce a higher risk to the internal campus. To protect the Campus Core from threats, the switches in the Edge Distribution module must protect the campus from the following attacks:

- **Unauthorized access:** All connections from the Edge Distribution module that pass through the Campus Core must be verified against the user and the user's rights. Filtering mechanisms must provide granular control over specific edge subnets and their capability to reach areas within the campus.
- **IP spoofing:**IP spoofing is a hacker technique for impersonating the identity of another user by using that user's IP address. Denial of service (DoS) attacks use IP spoofing to generate requests to servers, using the stolen IP address as a source. The server therefore does not respond to the original source, but it does respond to the stolen IP address. A significant amount of this type of traffic causes the attacked server to be unavailable, thereby interrupting business. DoS attacks are a problem because they are difficult to detect and defend against; attackers can use a valid internal IP address for the source address of IP packets that produce the attack.
- **Network reconnaissance:** Network reconnaissance (or discovery) sends packets into the network and collects responses from the network devices. These responses provide basic information about the internal network topology. Network intruders use this approach to find out about network devices and the services that run on them.

  Therefore, filtering traffic from network reconnaissance mechanisms before it enters the enterprise network can be crucial. Traffic that is not essential must be limited to prevent a hacker from performing network reconnaissance.

- **Packet sniffers:** Packet sniffers are devices that monitor and capture the traffic in the network and might be used by hackers. Packets belonging to the same broadcast domain are vulnerable to capture by packet sniffers, especially if the packets are broadcast or multicast. Because most of the traffic to and from the Edge Distribution module is business-critical, corporations cannot afford this type of security lapse. Multilayer switches can prevent such an occurrence.

The Edge Distribution devices provide the last line of defense for all external traffic that is destined for the Campus Infrastructure module. In terms of overall functionality, the Edge Distribution switches are similar to the Building Distribution layer switches. Both use access control to filter traffic, although the Edge Distribution switches can rely on the Enterprise Edge modules to provide additional security. Both modules use multilayer switching to achieve high performance, but the Edge Distribution module can provide additional security functions because its performance requirements might not be as high.

When the enterprise includes a significant data center rather than a simple server farm, remote connectivity and performance requirements are more stringent. Edge Distribution switches can be located in the data center, giving remote users easier access to corporate resources. Appropriate security concerns need to be addressed in this module.

Server Placement

Within a campus network, servers may be placed locally in the Building Access or Building Distribution layer, or attached directly to the Campus Core. Centralized servers are typically grouped into a server farm located in the Enterprise Campus or in a separate data center.

Servers Directly Attached to Building Access or Building Distribution Layer Switches

If a server is local to a certain workgroup that corresponds to one VLAN, and all workgroup members and the server are attached to a Building Access layer switch, most of the traffic to the server is local to the workgroup. If required, an access list at the Building Distribution layer switch could hide these servers from the enterprise.

In some midsize networks, building-level servers that communicate with clients in different VLANs, but that are still within the same physical building, can be connected to Building Distribution layer switches.

Servers Directly Attached to the Campus Core

The Campus Core generally transports traffic quickly, without any limitations. Servers in a medium-sized campus can be connected directly to Campus Core switches, making the servers closer to the users than if the servers were in a Server Farm, as illustrated in Figure 4-16. However, ports are typically limited in the Campus Core switches. Policy-based control (QoS and access control lists [ACL]) for accessing the servers is implemented in the Building Distribution layer, rather than in the Campus Core.

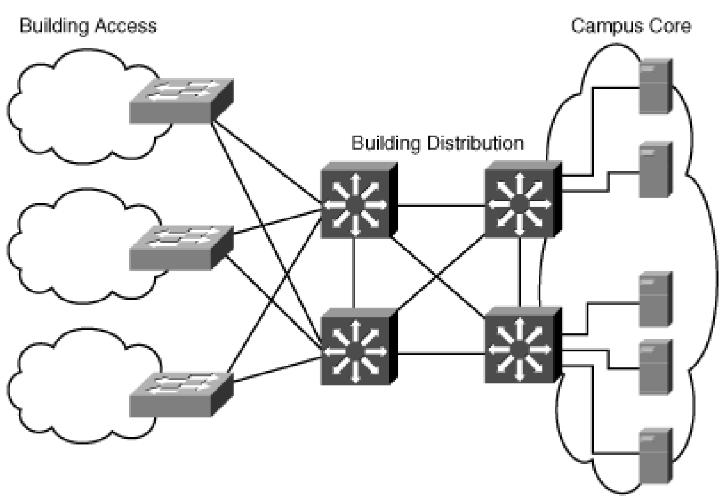

Figure 4-16: Servers Directly Attached to the Campus Core in a Medium-Sized Network

Servers in a Server Farm Module

Larger enterprises may have moderate or large server deployments. For enterprises with moderate server requirements, common servers are located in a separate Server Farm module connected to the Campus Core layer using multilayer server distribution switches, as illustrated in Figure 4-17. Because of high traffic load, the servers are usually Gigabit Ethernet–attached to the Server Farm switches. Access lists at the Server Farm module's multilayer distribution switches implement the controlled access to these servers. Redundant distribution switches in a Server Farm module and solutions such as the HSRP and GLBP provide fast failover. The Server Farm module distribution switches also keep all server-to-server traffic off the Campus Core.

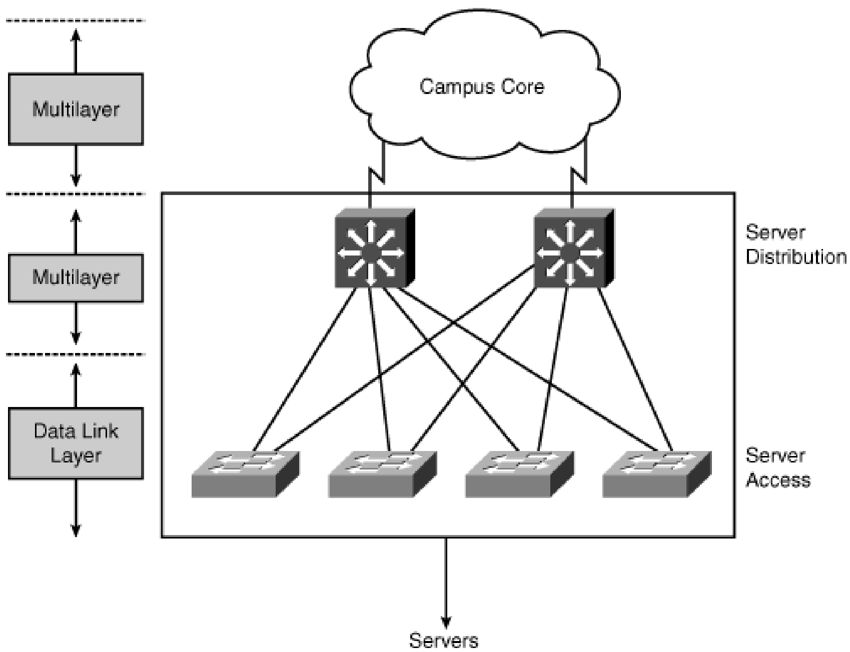

Figure 4-17: Server Farm in a Large Network

Rather than being installed on only one server, modern applications are distributed among several servers. This approach improves application availability and responsiveness. Therefore, placing servers in a common group (in the Server Farm module) and using intelligent multilayer switches provide the applications and servers with the required scalability, availability, responsiveness, throughput, and security.

For a large enterprise with a significant number of servers, a separate data center, possibly in a remote location, is often implemented. Design considerations for an Enterprise Data Center are discussed in the later "Enterprise Data Center Design Considerations" section.

Server Farm Design Guidelines

As shown in Figure 4-18, the Server Farm can be implemented as a high-capacity building block attached to the Campus Core using a modular design approach. One of the main concerns with the Server Farm module is that it receives the majority of the traffic from the entire campus. Random frame drops can result because the uplink ports on switches are frequently oversubscribed. To guarantee that no random frame drops occur for business-critical applications, the network designer should apply QoS mechanisms to the server links.

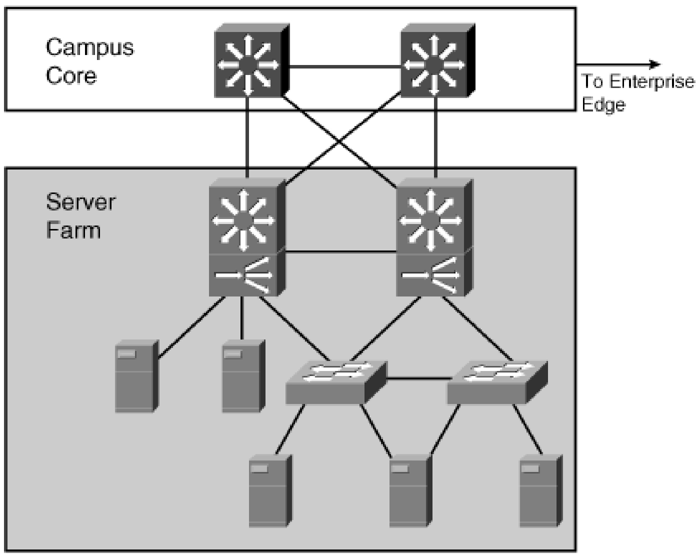

Figure 4-18: Sample Server Farm Design

Note*Switch oversubscription* occurs when a switch allows more ports (bandwidth) in the chassis than the switch's hardware can transfer through its internal structure.

The Server Farm design should ensure that the Server Farm uplink ports are not as oversubscribed as the uplink ports on the switches in the Building Access or Building Distribution layers. For example, if the campus consists of a few Building Distribution layers connected to the Campus Core layer with Gigabit Ethernet, attach the Server Farm module to the Campus Core layer with either a 10-Gigabit Ethernet or multiple Gigabit Ethernet links.

The switch performance and the bandwidth of the links from the Server Farm to the Campus Core are not the only considerations. You must also evaluate the server's capabilities. Although server manufacturers support a variety of NIC connection rates (such as Gigabit Ethernet), the underlying network operating system might not be able to transmit at the maximum line capacity. As such, oversubscription ratios can be raised, reducing the Server Farm's overall cost.

Server Connectivity Options

Servers can be connected in several different ways. For example, a server can attach by one or two Fast Ethernet connections. If the server is dual-attached (dual-NIC redundancy), one interface can be active while the other is in hot standby. Installing multiple single-port NICs or multiport NICs in the servers extends dual homing past the Server Farm module switches to the server itself. Servers needing redundancy can be connected with dual-NIC homing in the access layer or a NIC that supports EtherChannel. With the dual-homing NIC, a VLAN or trunk is needed between the two access switches to support the single IP address on the two server links to two separate switches.

Within the Server Farm module, multiple VLANs can be used to create multiple policy domains as required. If one particular server has a unique access policy, a unique VLAN and subnet can be created for that server. If a group of servers has a common access policy, the entire group can be placed in a common VLAN and subnet. ACLs can be applied on the interfaces of the multilayer switches.

Several other solutions are available to improve server responsiveness and evenly distribute the load to them. For example, Figure 4-18 includes content switches that provide a robust front end for the Server Farm by performing functions such as load balancing of user requests across the Server Farm to achieve optimal performance, scalability, and content availability.

The Effect of Applications on Switch Performance

Server Farm design requires that you consider the average frequency at which packets are generated and the packets' average size. These parameters are based on the enterprise applications' traffic patterns and number of users of the applications.

Interactive applications, such as conferencing, tend to generate high packet rates with small packet sizes. In terms of application bandwidth, the packets-per-second limitation of the multilayer switches might be more critical than the throughput (in Mbps). In contrast, applications that involve large movements of data, such as file repositories, transmit a high percentage of full-length (large) packets. For these applications, uplink bandwidth and oversubscription ratios become key factors in the overall design. Actual switching capacities and bandwidths vary based on the mix of applications.